Variable Selection for Wasserstein Robust Hypothesis Testing

1 Problem formulation

Throughout this paper, we consider the variable selection problem for robust hypothesis testing. Let $[D] = \{1, \ldots, D\}$ and d be a positive integer denoting the sparsity budget. Given a detector $f : \mathbb{R}^D \to \mathbb{R}$, we define the worst-case risk as

$$R(f) = \sup_{\mathbb{P}_0 \in \mathcal{P}_0, \mathbb{P}_1 \in \mathcal{P}_1} \left\{ \mathbb{E}_{\mathbb{P}_0} \ell(-f(X)) + \mathbb{E}_{\mathbb{P}_1} \ell(f(X)) \right\}, \tag{1}$$

where $\mathcal{P}_k, k = 0, 1$ denote the Wasserstein distance-based ambiguity sets, and $\ell(\cdot)$ is the generating function that is nonnegative-valued, non-decreasing, convex, and satisfies $\ell(0) = 1, \lim_{t \to -\infty} \ell(t) = 0$. For $k = 0, 1$, assume we have n_k available observations in the group k, which are denoted as $\{x_i^{(k)}\}_{i=1}^{n_k}$. We denote the corresponding empirical distribution as $\widehat{\mathbb{P}}_k = \frac{1}{n_k} \sum_{i=1}^{n_k} \delta_{x_i^{(k)}}$, and construct the Wasserstein ambiguity set as

$$\mathcal{P}_k = \left\{ \mathbb{P}_k : \mathcal{W}_1(\widehat{\mathbb{P}}_k, \mathbb{P}_k) \le \rho_k \right\}.$$

Here, $\mathcal{W}_1(\cdot, \cdot)$ refers to the 1-Wasserstein distance equipped with the ℓ_1-norm.

For $S \subseteq [D]$, define the collection of detectors whose input only depends on coordinates belonging to S:

$$\mathcal{F}_S = \left\{ f : \mathbb{R}^D \to \mathbb{R}, f(x) = g(x[S]) \text{ for some measurable function } g : \mathbb{R}^{|S|} \to \mathbb{R} \right\}.$$

Note that for a given vector $x \in \mathbb{R}^D$, we use $x[i]$ to denote its i-th coordinate, and $x[S]$ to denote the concatenation of its coordinates whose indices belong to S. Our proposed variable selection problem can be formulated as the following combinatorial optimization problem

$$\min_{S \subseteq [D], |S| \le d} \min_{f \in \mathcal{F}_S} R(f). \tag{2}$$

2 Convex formulation

For $S \subseteq [D]$, denote $V(S) = \min_{f \in \mathcal{F}_S} R(f)$ and define

$$\mathcal{X} = \{x_i^{(0)} : i \in [n_0]\} \cup \{x_j^{(1)} : j \in [n_1]\}.$$

Assume $n_0, n_1 \ge 1$, $\rho_0, \rho_1 > 0$, and that ℓ is finite on $\mathbb{R}$. Each target $y \in \mathcal{X}$ receives a detector score g_y, and $g = (g_y)_{y \in \mathcal{X}}$. The next lemma is [2, Proposition 1 and Theorem 3] applied to the projected observations, in the transport notation of [4]: projection contracts the ℓ_1 transport cost, and every projected transport lifts at the same cost by keeping each source's unselected coordinates.

Lemma 2.1 (Least favorable distributions)**.**

$$\begin{aligned}
V(S) = \max_{p, \gamma \ge 0} \quad & \sum_{y \in \mathcal{X}} \inf_{g_y \in \mathbb{R}} \{p_0(y)\ell(-g_y) + p_1(y)\ell(g_y)\} \\
\textit{subject to} \quad & \sum_{y \in \mathcal{X}} \gamma_{i,y}^{(k)} = \frac{1}{n_k}, && i \in [n_k],\ k = 0, 1, \\
& p_k(y) = \sum_{i=1}^{n_k} \gamma_{i,y}^{(k)}, && y \in \mathcal{X},\ k = 0, 1, \\
& \sum_{i=1}^{n_k} \sum_{y \in \mathcal{X}} \gamma_{i,y}^{(k)} \|x_i^{(k)}[S] - y[S]\|_1 \le \rho_k, && k = 0, 1.
\end{aligned} \tag{3}$$

Here $\gamma_{i,y}^{(k)}$ is the mass transported from $x_i^{(k)}[S]$ to $y[S]$, and $p_k(y)$ is the mass of class k at that target. Let Γ_S denote the feasible transports in Eq. (3) after eliminating p. Since Γ_S is compact and convex, and the sum below is convex in g and linear in γ, Sion's minimax theorem [3] gives

$$\begin{aligned} V(S) &= \max_{\gamma\in\Gamma_S}\inf_{g}\sum_{k=0}^{1}\sum_{i,y}\gamma_{i,y}^{(k)}\ell((-1)^{k+1}g_y) \\ &= \inf_{g}\max_{\gamma\in\Gamma_S}\sum_{k=0}^{1}\sum_{i,y}\gamma_{i,y}^{(k)}\ell((-1)^{k+1}g_y). \end{aligned}$$

For fixed g, the inner maximum is a linear program in each class, feasible because leaving every observation in place costs nothing. Its Lagrangian is

$$\begin{aligned} \mathcal{L}_k = &\sum_{i,y}\gamma_{i,y}^{(k)}\ell((-1)^{k+1}g_y) + \sum_i a_i^{(k)}\left(\frac{1}{n_k} - \sum_y \gamma_{i,y}^{(k)}\right) \\ &+ \lambda_k\left(\rho_k - \sum_{i,y}\gamma_{i,y}^{(k)}\|x_i^{(k)}[S] - y[S]\|_1\right), \end{aligned}$$

with multipliers $a_i^{(k)} \in \mathbb{R}$ for source-mass conservation and $\lambda_k \geq 0$ for the transport budget. Strong duality, followed by minimization over g, gives

$$\begin{aligned} V(S) = \min_{g,a,\lambda}\quad & \frac{1}{n_0}\sum_i a_i^{(0)} + \frac{1}{n_1}\sum_j a_j^{(1)} + \rho_0\lambda_0 + \rho_1\lambda_1 \\ \text{subject to}\quad & \ell(-g_y) \leq a_i^{(0)} + \lambda_0\|x_i^{(0)}[S] - y[S]\|_1, \qquad i\in[n_0],\ y\in\mathcal{X}, \\ & \ell(g_y) \leq a_j^{(1)} + \lambda_1\|x_j^{(1)}[S] - y[S]\|_1, \qquad j\in[n_1],\ y\in\mathcal{X}, \\ & g\in\mathbb{R}^{\mathcal{X}},\quad a^{(k)}\in\mathbb{R}^{n_k},\quad \lambda_k\geq 0, \qquad k=0,1. \end{aligned} \tag{4}$$

Eq. (4) is a finite convex program, and it attains its minimum: the constraints with $y = x_i^{(k)}$ give $a_i^{(k)} \geq 0$, so an objective value at most B forces $a_i^{(k)} \leq n_k B$ and $\lambda_k \leq B/\rho_k$; the constraints then bound $\ell(\pm g_y)$, hence g, since $\ell(t) \to \infty$ as $t \to \infty$ (ℓ is convex with $\ell(0) = 1 > \lim_{t\to-\infty}\ell(t)$).

Following [2, 4], a detector is recovered from an optimal (g, a, λ) as follows. Coincident projected targets have identical constraints, so set $g(y[S]) = g_y$. For any other $\xi \in \mathbb{R}^{|S|}$, let i and j be the smallest indices minimizing $a_r^{(0)} + \lambda_0\|x_r^{(0)}[S] - \xi\|_1$ over $r \in [n_0]$ and $a_r^{(1)} + \lambda_1\|x_r^{(1)}[S] - \xi\|_1$ over $r \in [n_1]$, respectively. With $r_0 = \|x_i^{(0)}[S] - \xi\|_1$ and $b = \|x_i^{(0)}[S] - x_j^{(1)}[S]\|_1$, define

$$\tau = \begin{cases} \min\{1, r_0/b\}, & b > 0, \\ 0, & b = 0, \end{cases} \qquad g(\xi) = (1-\tau)g_{x_i^{(0)}} + \tau g_{x_j^{(1)}}. \tag{5}$$

Convexity of ℓ and the triangle inequality bound $\ell(-g(\xi))$ by $a_i^{(0)} + \lambda_0 r_0$, and $\ell(g(\xi))$ by the analogous quantity at j; by the choice of i and j, the constraints of Eq. (4) therefore hold at every $\xi \in \mathbb{R}^{|S|}$. Integrating them against couplings within the budgets shows that the Borel detector $f(x) = g(x[S])$ attains $R(f) = V(S)$.

To optimize the coordinate selection, define $z = \mathbf{1}_S$ and $\mathcal{Z}_d = \{z \in \{0,1\}^D : \mathbf{1}^\top z \leq d\}$. For $\gamma = (\gamma^{(0)}, \gamma^{(1)})$ with $\gamma^{(k)} \in \mathbb{R}_+^{n_k\times|\mathcal{X}|}$, define the minimum weighted loss over the finite scores in Eq. (4):

$$H(\gamma) = \inf_{g\in\mathbb{R}^{\mathcal{X}}}\sum_{y\in\mathcal{X}}\left[\sum_{i=1}^{n_0}\gamma_{i,y}^{(0)}\ell(-g_y) + \sum_{j=1}^{n_1}\gamma_{j,y}^{(1)}\ell(g_y)\right].$$

Define the average of the two class masses and the transport cost in coordinate m, respectively, by

$$M(\gamma) = \frac{1}{2}\sum_{k=0}^{1}\sum_{i=1}^{n_k}\sum_{y\in\mathcal{X}}\gamma_{i,y}^{(k)},$$
$$c_k(\gamma)[m] = \sum_{i=1}^{n_k}\sum_{y\in\mathcal{X}}\gamma_{i,y}^{(k)}|x_i^{(k)}[m] - y[m]|, \qquad k = 0,1,\ m \in [D].$$

For a scalar $t \in \mathbb{R}$, write $t_+ = \max\{t, 0\}$. Following [1], the next result expresses the selection objective as a maximum of functions affine in z.

Proposition 2.2 (Convex selection function)**.** *For $z \in [0,1]^D$, define*

$$\begin{aligned} F(z) = \max_{\gamma\geq 0,\ \mu\in\mathbb{R}_+^D}\ & 2\sqrt{H(\gamma)} - M(\gamma) - \mu^\top z \\ \text{subject to}\ & \sum_{y\in\mathcal{X}}\gamma_{i,y}^{(k)} = \frac{M(\gamma)}{n_k}, && i \in [n_k],\ k = 0,1, \\ & \sum_{m=1}^{D}\big(c_k(\gamma)[m] - \rho_k\mu_m\big)_+ \leq \rho_k M(\gamma), && k = 0,1. \end{aligned} \tag{6}$$

Then F is convex and nonincreasing, the maximum in Eq. (6) is attained, and

$$F(\mathbf{1}_S) = V(S), \qquad \min_{|S|\leq d} V(S) = \min_{z\in\mathcal{Z}_d} F(z). \tag{7}$$

The row constraints make each class's total mass equal to $M(\gamma)$; this common mass is allowed to vary, whereas each class has unit mass in Eq. (3). Appendix A derives Eq. (6) from the epigraph of Eq. (4) by duality and a change of scale. Minimizing F over $\mathcal{Z}_d$ therefore solves the original problem; replacing $\mathcal{Z}_d$ by its continuous relaxation gives a lower bound.

3 An outer-approximation approach

We use an outer-approximation approach following [1], which iteratively refines affine lower bounds on F over $\mathcal{Z}_d$. Since H is nonnegative and concave, Eq. (6) is a convex optimization problem for fixed z. For an optimizer (γ^t, μ^t) at z^t, define

$$\begin{aligned} c_t(z) &= 2\sqrt{H(\gamma^t)} - M(\gamma^t) - (\mu^t)^\top z \\ &= F(z^t) - (\mu^t)^\top(z - z^t). \end{aligned} \tag{8}$$

The feasible set in Eq. (6) is independent of z, so $c_t(z) \leq F(z)$ for every $z \in [0,1]^D$, with equality at z^t. After t evaluations, we minimize the accumulated lower bounds to obtain the next selection:

$$\begin{aligned} (z^{t+1}, L_t) \in \operatorname*{arg\,min}_{z\in\mathcal{Z}_d,\ \eta\geq 0}\ & \eta \\ \text{subject to}\ & \eta \geq c_s(z), \qquad s = 1,\dots,t. \end{aligned} \tag{9}$$

The value L_t is a lower bound on $\mathrm{OPT} = \min_{z\in\mathcal{Z}_d} F(z)$, and the best evaluated value $U_t = \min_{s\leq t} F(z^s)$ is an upper bound.

Algorithm 1 alternates between evaluating a selection and solving Eq. (9). It retains the best evaluated detector and stops when $U_t - L_t \leq \varepsilon$. With exact solves, if $z^{t+1} = z^s$ for some $s \leq t$, then

$$L_t \geq c_s(z^{t+1}) = F(z^s) \geq U_t,$$

so the stopping condition holds. Each nonterminal iteration therefore produces a new selection, and at most $|\mathcal{Z}_d|$ selections are evaluated. Since F is nonincreasing, the budget can also be imposed as $\mathbf{1}^\top z = d$ when $d \leq D$.

When a solver stops early, feasible solutions of Eq. (4) give upper bounds, and feasible solutions of Eq. (6) give cuts $2\sqrt{H(\gamma)} - M(\gamma) - \mu^\top z$. Use the solver's global lower bound for Eq. (9) as L_t.

Algorithm 1 Outer approximation for variable selection

Require: Samples $x_i^{(k)}$, radii $\rho_k > 0$, budget d, tolerance $\varepsilon > 0$, initial $z^1 \in \mathcal{Z}_d$

$U \leftarrow +\infty$, $t \leftarrow 1$
loop
  $S^t \leftarrow \{m : z_m^t = 1\}$
  Solve Eq. (4) at S^t for (g^t, a^t, λ^t) and value $v_t = V(S^t)$
  Solve Eq. (6) at z^t for (γ^t, μ^t); form c_t by Eq. (8)
  if $v_t < U$ **then**
    $U \leftarrow v_t$; store $S^\star \leftarrow S^t$ and $(g^\star, a^\star, \lambda^\star) \leftarrow (g^t, a^t, \lambda^t)$
  end if
  Solve Eq. (9) with $c_1, \ldots, c_t$ for (z^{t+1}, L_t)
  if $U - L_t \le \varepsilon$ **then**
    Recover $g^\star$ on $\mathbb{R}^{|S^\star|}$ by Eq. (5)
    return $S^\star$ and $f^\star(x) = g^\star(x[S^\star])$
  end if
  $t \leftarrow t + 1$
end loop

A Proof of Proposition 2.2

A.1 Duality

Let $\mathcal{D}$ denote the feasible set in Eq. (6). Introduce an epigraph variable η in Eq. (4) and write $\|x_i^{(k)}[S] - y[S]\|_1 = \sum_m z_m |x_i^{(k)}[m] - y[m]|$ for $z = \mathbf{1}_S$. The diagonal constraints give $a_i^{(k)} \ge 0$, so the epigraph inequality implies $\sum_k \rho_k \lambda_k \le \eta$. Introducing w_{km} for $\lambda_k z_m$, consider the following problem for fixed $z \in [0,1]^D$:

$$
\begin{aligned}
v(z) = \min_{g,a,\lambda,w,\eta} \quad & \eta \\
\text{subject to} \quad & \ell((-1)^{k+1} g_y) \le a_i^{(k)} + \sum_m w_{km} |x_i^{(k)}[m] - y[m]|, && k = 0,1,\ i \in [n_k],\ y \in \mathcal{X}, \\
& 0 \le w_{km} \le \lambda_k, && k = 0,1,\ m \in [D], \\
& \sum_{k=0}^{1} \rho_k w_{km} \le \eta z_m, && m \in [D], \\
& \sum_{k=0}^{1} \left(\frac{1}{n_k} \sum_i a_i^{(k)} + \rho_k \lambda_k \right) \le \eta, \\
& g \in \mathbb{R}^{\mathcal{X}}, \quad a^{(k)} \in \mathbb{R}^{n_k}, \quad \eta \in \mathbb{R}.
\end{aligned}
\tag{10}
$$

Using nonnegative Lagrange multipliers $\gamma, \beta, \mu, \alpha$ for the loss, weight, coordinate, and epigraph inequalities, respectively, we obtain

$$
\begin{aligned}
\mathcal{L} = (1 - \alpha - \mu^\top z)\eta &+ \sum_{k,i,y} \gamma_{i,y}^{(k)} \ell((-1)^{k+1} g_y) \\
&+ \sum_{k,i} a_i^{(k)} \left(\frac{\alpha}{n_k} - \sum_y \gamma_{i,y}^{(k)} \right) + \sum_k \lambda_k \left(\rho_k \alpha - \sum_m \beta_{km} \right) \\
&+ \sum_{k,m} w_{km} \left[\beta_{km} + \rho_k \mu_m - c_k(\gamma)[m] \right].
\end{aligned}
$$

Minimization over g gives $H(\gamma)$. Minimization over the free variables a, η and the nonnegative variables λ, w is finite precisely when

$$\sum_y \gamma_{i,y}^{(k)} = \alpha/n_k, \qquad \alpha + \mu^\top z = 1,$$
$$\sum_m \beta_{km} \le \rho_k \alpha, \qquad \beta_{km} \ge c_k(\gamma)[m] - \rho_k \mu_m, \qquad \beta_{km} \ge 0.$$

Summing the row equations over i shows that each class has total mass α, so $\alpha = M(\gamma)$. The smallest admissible multiplier is $\beta_{km} = (c_k(\gamma)[m] - \rho_k \mu_m)_+$, so a suitable β exists exactly when the last constraint of Eq. (6) holds. Eliminating α and β thus leaves $(\gamma, \mu) \in \mathcal{D}$ and $M(\gamma) + \mu^\top z = 1$.

Problem Eq. (10) is feasible (take $w = 0$ and a feasible point of Eq. (4) with $S = \emptyset$), and its minimum is attained by the argument given for Eq. (4), since $0 \le w_{km} \le \lambda_k$. Adding $\delta > 0$ to every $a_i^{(k)}$ and 3δ to η makes the loss constraints strict and keeps the affine ones, so Slater's condition gives strong duality with an attained dual optimum:

$$v(z) = \max_{\substack{(\gamma,\mu)\in\mathcal{D} \\ M(\gamma)+\mu^\top z = 1}} H(\gamma). \tag{11}$$

A.2 Equality at binary selections

We prove $v(\mathbf{1}_S) = V(S)$. Let $\gamma^\star$ maximize Eq. (3) and set

$$\mu_m = \begin{cases} 0, & m \in S, \\ \max\limits_{k=0,1} \dfrac{c_k(\gamma^\star)[m]}{\rho_k}, & m \notin S. \end{cases}$$

Then $M(\gamma^\star) = 1$, $\mu^\top \mathbf{1}_S = 0$, and

$$\sum_m \big(c_k(\gamma^\star)[m] - \rho_k \mu_m\big)_+ = \sum_{m\in S} c_k(\gamma^\star)[m] \le \rho_k.$$

Thus $(\gamma^\star, \mu)$ is feasible in Eq. (11), giving $v(\mathbf{1}_S) \ge V(S)$.

For the reverse inequality, let (γ, μ) be feasible in Eq. (11) at $z = \mathbf{1}_S$. The normalization gives $0 \le M(\gamma) \le 1$. Let $\iota_{i,y}^{(k)} = n_k^{-1} \mathbf{1}_{\{y = x_i^{(k)}\}}$ be the transport that leaves each observation in place, so $c_k(\iota)[m] = 0$ for all k, m. Completing the missing mass gives $\gamma' = \gamma + (1 - M(\gamma))\iota$, whose rows sum to $1/n_k$ and whose cost satisfies

$$\begin{aligned} \sum_{m\in S} c_k(\gamma')[m] = \sum_{m\in S} c_k(\gamma)[m] &\le \sum_m \big(c_k(\gamma)[m] - \rho_k \mu_m\big)_+ + \rho_k \sum_{m\in S} \mu_m \\ &\le \rho_k \left(M(\gamma) + \mu^\top \mathbf{1}_S\right) = \rho_k. \end{aligned}$$

Hence $\gamma' \in \Gamma_S$, and since $\gamma' \ge \gamma$ and $\ell \ge 0$, $V(S) \ge H(\gamma') \ge H(\gamma)$. Maximizing over Eq. (11) proves $V(S) \ge v(\mathbf{1}_S)$.

A.3 Scaling and convexity

It remains to remove the normalization. The constraints defining $\mathcal{D}$ are homogeneous in (γ, μ), and $H(s\gamma) = sH(\gamma)$ for $s \ge 0$. If $M(\gamma) + \mu^\top z = 0$, the row constraints force $\gamma = 0$, and the objective in Eq. (6) is zero. Otherwise, division by $M(\gamma) + \mu^\top z$ produces a feasible pair in Eq. (11). Since

$$2\sqrt{sh} - s = h - (\sqrt{s} - \sqrt{h})^2, \qquad \max_{s\ge 0}\{2\sqrt{sh} - s\} = h \quad (h \ge 0), \tag{12}$$

we obtain

$$\begin{aligned} F(z) &= \max_{\substack{(\gamma,\mu)\in\mathcal{D} \\ M(\gamma)+\mu^\top z = 1}} \max_{s\ge 0}\{2\sqrt{sH(\gamma)} - s\} \\ &= \max_{\substack{(\gamma,\mu)\in\mathcal{D} \\ M(\gamma)+\mu^\top z = 1}} H(\gamma) = v(z). \end{aligned}$$

The scalar maximum is attained at $s = h$, so attainment in Eq. (11) also gives attainment in Eq. (6). In particular, $F(\mathbf{1}_S) = V(S)$.

Finally, $\mathcal{D}$ is convex, $\sqrt{H}$ is concave, and M is linear. Hence Eq. (6) is a convex optimization problem for fixed z. Its objective is affine in z with slope $-\mu \leq 0$, and its feasible set is independent of z, so F is convex and nonincreasing. The correspondence $S \leftrightarrow \mathbf{1}_S$ gives Eq. (7). □

References

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